

AHMED REFAT

Data Engineer

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<https://github.com/AhmedRefat1412> | [Portfolio](#)

PROFESSIONAL SUMMARY

Results-driven Data Engineer with proven experience architecting and delivering end-to-end batch and real-time streaming data pipelines, cloud-native data warehouses, and production-grade Lakehouse systems. Proficient in Python, PySpark, Apache Kafka, and Apache Airflow, with hands-on expertise across AWS and Azure cloud platforms. Skilled in Medallion Architecture (Bronze/Silver/Gold), Apache Iceberg, Snowflake, and dbt. Demonstrates a consistent ability to transform raw, high-volume datasets into reliable, analytics-ready assets that drive business intelligence and machine learning at scale.

TECHNICAL SKILLS

Languages & Processing: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, Pyspark

Big Data & Streaming: Apache Spark, Apache Kafka, Apache Iceberg, Hadoop / HDFS

Cloud Platforms: AWS (S3, Glue, Redshift, IAM), Azure (ADLS, Data Factory, Synapse Analytics), Snowflake, Databricks

Data Warehousing & Modeling: Dimensional Modeling, Star/Galaxy Schema, dbt, Snowflake, SQL Server, PostgreSQL, MongoDB

Orchestration & ETL: Apache Airflow, dbt, ETL/ELT Pipeline Design, Great Expectations (Data Quality)

DevOps & Infrastructure: Docker, Git / GitHub, Linux, Bash

BI & Monitoring: Power BI, Grafana

Architecture Patterns: Medallion Architecture (Bronze/Silver/Gold), Lakehouse, Data Warehouse, Lambda Architecture

EXPERIENCE & TRAINING

Data Engineer Trainee | Microsoft - Digital Egypt Pioneer Initiative (DEPI)

Nov 2025 – May 2026

- Designed and implemented data pipelines using Python and SQL across 2+ cloud-based projects, applying ETL/ELT patterns to transform raw data into analytics-ready datasets.
- Applied dimensional data modeling techniques including Star and Galaxy Schemas, improving data structure consistency and reducing analytical query complexity.
- Completed a 6-month data engineering program covering the full pipeline lifecycle, ingestion, transformation, warehousing, orchestration and Azure cloud infrastructure.

Cloud Services Trainee | National Telecommunication Institute (NTI)

Jan 2026 – Mar 2026

- Deployed and configured AWS cloud services across a 2-month hands-on program, including EC2 provisioning, IAM policy configuration, Redshift, Glue, RDS, Dynamodb, and S3 storage management across multiple lab environments.
- Practiced Linux system administration and network fundamentals, strengthening operational skills for cloud-native data engineering environments.

PROJECTS

Financial Data Platform: AWS Cloud Data Warehouse | [GitHub](#)

- Built a Medallion Architecture (Bronze → Silver → Gold) ELT pipeline on AWS processing 2M+ financial transaction records, enabling fraud detection analytics through structured data transformation.
- Implemented a data quality gate using Great Expectations before Silver layer processing, validating schema, completeness, and accuracy of raw financial data ingested from S3.
- Processed and transformed data using AWS Glue (PySpark), loaded into Amazon Redshift with a Star Schema design, and applied dbt for business logic modeling and dataset standardization.
- Delivered a Power BI fraud detection dashboard that gets the data from Redshift, orchestrated via Apache Airflow.

NYC Smart Traffic: Batch & Streaming Analytics Platform | [GitHub](#)

- Built 2 fully independent pipelines: a batch pipeline ingesting NYC traffic, weather, and incident data from NYC Open Data into a 3-layer S3 data lake (Raw → Processed → Curated) using PySpark, orchestrated by Apache Airflow. And a separate streaming pipeline consuming live Citi Bike station data from a real-time API into Kafka topics processed by PySpark Structured Streaming.
- Batch pipeline output: modeled a Galaxy Schema data warehouse in Snowflake covering traffic, weather, and incident domains (40M+ records), delivering historical business insights via Power BI dashboards.
- Streaming pipeline output: stored processed Citi Bike station data in TimescaleDB for time-series querying, with a live Grafana dashboard for real-time station availability and usage monitoring.

Predictive Telecom Streaming Lakehouse | [GitHub](#)

- Building a production-grade streaming Data Lakehouse on Apache Iceberg / AWS S3, ingesting 4 simulated API streams (system, network, radio, environmental metrics) into Kafka topics for real-time processing.
- Processing streams through a Bronze → Silver → Gold Medallion Architecture using PySpark Structured Streaming. Silver layer joins all 4 streams into unified records stored as Iceberg tables on S3 with ACID support and automated compaction.
- Applied an Isolation Forest model on each Gold layer micro-batch for real-time anomaly detection; built a fully independent Python microservice consuming a separate Kafka topic, applying custom threshold conditions to trigger Telegram Bot alerts without coupling to the main ML pipeline.
- Orchestrated 2 Airflow DAGs - one for periodic Iceberg compaction across all 3 layers, one for Isolation Forest model retraining on historical lakehouse data . with live monitoring via Grafana connected to AWS Athena over the Gold layer.

Flight Delay Prediction Platform | [GitHub](#)

- Built a unified Streaming + Batch platform integrating 2 live APIs (OpenSky for flight data, Open-Meteo for weather); streaming path sends data through Kafka → PySpark Structured Streaming → TimescaleDB → Grafana for real-time monitoring.
- Integrated an XGBoost Classifier within the streaming pipeline for real-time flight delay prediction (82% accuracy), applied on each micro-batch to deliver live predictions alongside raw telemetry.
- Built a batch pipeline using Medallion Architecture on S3 + PySpark (6-step Airflow DAG: Bronze → Silver → Gold → Snowflake), modeled a Galaxy Schema data warehouse in Snowflake, and delivered 2 Power BI analytical dashboards for historical delay pattern analysis.

EDUCATION

B.Sc. Computer Science & Information Systems

Graduated: May 2023

Information Systems Department
Kafr El-Sheikh University, Faculty of Computers and Information

CERTIFICATIONS

Microsoft Data Engineer — DEPI	May 2026
AWS Certified Solutions Architect — Associate	Apr 2026
AWS Cloud Foundations — AWS	Apr 2026
Cloud Services Management and Operation — NTI	Mar 2026
Linux Command Line & Docker — Udemy	Jan 2026
Analyzing and Visualizing Data with Power BI — Udemy	Dec 2025
Introduction to MongoDB — MongoDB University	Sep 2025

LANGUAGES

Arabic: Native | **English:** Professional Working Proficiency (B2+)